

Cloud Architecture for video streaming

* A media Company wants to support millions of concurrent users with minimal buffering

Proposed Architecture :-

Users $\rightarrow$ CDN $\rightarrow$ Load Balancer $\rightarrow$ Auto-Scaling Servers $\rightarrow$ Cloud Storage

1. CDN: Caches videos at locations close to users, reducing latency and buffering
2. Cloud storage: stores large video files in scalable object storage
3. Load Balancer: Distributes user requests across multiple servers
4. Caching: frequently accessed videos are cached to reduce requests to the main servers.

Cost optimization :-

- * use auto-scaling to avoid unnecessary resources
- * Use CDN caching to reduce data transfer from the main servers
- * Monitor resource usage and remove unused resources

Conclusions:-

- * Using CDN, Caching, load balancing, and auto-scaling provide smooth and reliable streaming for millions of users while controlling cloud costs.

Container orchestration and multi-cloud strategy

- * A fintech startup is deploying applications using containers across multiple cloud providers.

Proposed solution

Container orchestration :-

- * Use a Container orchestration platform to manage Containers across different cloud providers
- * It automates deployment, scaling.

2) Service Discovery :-

- * A service discovery mechanism helps Containers find and communicate with other services automatically.

- * This avoids manually configuring service locations.

3) Scaling :-

- * Automatically increase Containers when workload increases.
- * Reduce Containers when demand decreases

Benefits :-

- * Avoids dependence on a single cloud provider
- * Improves availability and reliability.
- * Provides flexibility

Risks :-

- * Increased management complexity
- * higher monitoring and operational costs

Conclusion :-

- * Contains orchestration combined with a multi-cloud strategy provider consistent deployment, availability, efficient scaling, and better management is required to handle its complexity and risks.

Backup and Disaster Recovery - Strategy

Proposed solution :-

1. Cloud Backup :-

- * Take regular automated backups of important data.

- * stores backups separately so deleted data can be restored

2. Replication :-

- * Replicate important data to another cloud location

*) Versioning :-

- * Enable versioning to keep previous versions of files

- * Accidentally deleted or modified data can be recovered.

RTO and RPO :-

* RTO (Recovery Time Objective) : Defines how quickly the system must be restored after a failure

* RPO : Defines the maximum amount of recent data that can be lost.

Business Continuity steps :-

* Detect the failure or data loss.

* Restore data from the latest backup /

Version

* Switch to the replicated system

* verify data and application functionality

* Resume normal operations.

Conclusion :-

* Using backup, replication, and versioning provides reliable data recovery and helps the e-learning platform continue its services during failures.

Challenges :

- * Managing two different environments can be complex
- * Maintaining consistent security policies is difficult.
- * Compliance requirements must be continuously maintained.

Conclusion:

- * A hybrid cloud provides for sensitive workloads while allowing non-sensitive applications to benefit from cloud scalability and flexibility.

4) Hybrid cloud Architecture

* A government agency wants to use a hybrid cloud for sensitive and non-sensitive data.

Proposed Architecture:

On-Premises Data Center $\leftrightarrow$ Secure Connection $\leftrightarrow$ Public cloud.

1. On-Premises: Store sensitive data and applications that must follow compliance requirements.
2. Public cloud: Store and process non-sensitive data and applications.
3. Secure Communication: Use a secure private connection or VPN between on-premises and cloud environments.

Multi - Region Deployment Strategy:

* A SaaS Provider needs high availability and fault tolerance so that the application continues working even if one region goes down.

Proposed strategy:-

1. Multi - Region Deployment:-

* Deploy the application in two or more cloud regions

* If one region fails, users can be served from another region.

2. Data Replication:-

* Replicate important data between regions

* This ensures that updated data is available in the backup region

3. failover :-
~~~~~

- \* use health checks to detect regional failures
- \* Automatically redirect users to the healthy region.

4. Monitoring and Alerting :-

- \* Continuously monitor servers, applications, network traffic and errors.

Conclusion :-  
~~~~~

- * A multi-region Architecture with data replication and automatic failover provides high-availability and fault tolerance. Continuous monitoring and alerting help detect problems early and maintain reliable services.